

Laser-hybrid Accelerator for Radiobiological Applications (LhARA)

The LhARA collaboration

- 1 Arizona State University, Tempe, Arizona, US
- 2 Patrick G Johnston Centre for Cancer Research, Queens University Belfast, 97 Lisburn Road, Belfast, BT9 7AE, UK
- 3 School of Mathematics and Physics, Queen's University Belfast, University Road, Belfast, BT7 1NN, Northern Ireland, UK
- 4 Dept of Medical Physics and Biomedical Engineering, University College London, WC1E 6BT
- 5 Department of Medical Physics, University Hospital Birmingham Foundation NHS Trust, Edgbaston, Birmingham, B15 2TH, UK
- 6 Department of Cancer and Genomic Sciences, College of Medicine and Health, University of Birmingham, Edgbaston, Birmingham, B15 2TT, UK
- 7 School of Physics and Astronomy, University of Birmingham, Edgbaston, Birmingham, B15 2TT, UK
- 8 Department of Brain Sciences, Faculty of Medicine, Imperial College London, Exhibition Road, London SW7 2AZ, UK
- 9 CERN, CH-1211 Genève 23, Switzerland
- 10 Cockcroft Institute for Accelerator Science, Keckwick Ln, Daresbury, Warrington WA4 4AD
- 11 Department of Physics and Astronomy, The University of Manchester, Oxford Rd, Manchester, M13 9PL, UK
- 12 Institut Curie, Universit PSL, CNRS UMR3347, Inserm U1021, Signalisation Radiobiologie et Cancer, 91400 Orsay, France
- 13 Institut des Sciences Moléculaires d'Orsay, Rue André Rivière, Bâtiment 520, 91400 Orsay, France
- 14 Lawrence Berkeley National Laboratory, 1 Cyclotron Road, Berkeley, CA 94720, USA
- 15 Joint Department of Physics, Institute of Cancer Research and Royal Marsden NHS Foundation Trust, Sir Richard Doll Building, 28 Oakleaf Avenue, Sutton, London, SM2 5GP
- 16 INFN Catania, Via Santa Sofia, 64 - 95123 Catania - Italy
- 17 Department of Bioengineering, Imperial College London, Exhibition Road, London, SW7 2AZ, UK
- 18 Department of Aeronautics, Imperial College London, Exhibition Road, London, SW7 2AZ, UK
- 19 Department of Computing, Imperial College London, Exhibition Road, London, SW7 2AZ, UK
- 20 CRUK PPI group, Charing Cross Hospital, London, W6 8RF, UK
- 21 Dept. Radiation Physics and Radiobiology, Imperial College Healthcare NHS Trust, London, W2 1NY, UK
- 22 Imperial College NHS Healthcare Trust, Department of Clinical Oncology, Charing Cross Hospital, London W6 8RF, UK
- 23 Department of Physics, Imperial College London, Exhibition Road, London, SW7 2AZ, UK
- 24 Department of Surgery and Cancer, Imperial College, Hammersmith Hospital London, W12 0NN, UK
- 25 The John Adams Institute for Accelerator Science, University of Oxford, Keble Rd, Oxford, OX1 3RH
- 26 The John Adams Institute for Accelerator Science, Royal Holloway University of London
- 27 The John Adams Institute for Accelerator Science, Department of Physics, Imperial College London, Exhibition Road, London SW7 2AZ, UK
- 28 Fakultät für Physik, Lehrstuhl für Medizinische Physik, Ludwig-Maximilians-Universität München, Am Coulombwall 1, D-85748 Garching, Germany
- 29 Leo Cancer Care, Broadview, Windmill Hill, Hailsham, East Sussex, BN27 4RY, UK
- 30 Department of Physics, University of Liverpool, Liverpool, L69 7ZE, UK
- 31 National Physical Laboratory, Hampton Road, Teddington, TW11 0LW, UK
- 32 Department of Oncology, University of Oxford, Old Road Campus Research Building, Roosevelt Drive, Oxford, OX3 7DQ, UK
- 33 Particle Physics, Denys Wilkinson Building, Keble Rd, Oxford, OX1 3RH
- 34 Particle Physics Department, STFC Rutherford Appleton Laboratory, Harwell Oxford, Didcot OX11 0QX, UK
- 35 Department of Physics, Imperial College London, Exhibition Road, London SW7 2AZ, UK

- 36 *School of Physical and Chemical Sciences, Queen Mary University of London, Mile End Road, London, E1 4NS, UK*
37 *Rosalind Franklin Institute, Harwell Campus, Didcot, OX11 0QX, UK*
38 *Department of Physics, Royal Holloway University of London, Egham, Surrey, TW20 0EX, UK*
39 *Accelerator Science and Technology Centre, STFC Daresbury Laboratory, Daresbury, Cheshire, WA4 4AD, UK*
40 *Business Innovation Department, STFC Daresbury Laboratory, Daresbury, Cheshire, WA4 4AD, UK*
41 *Central Laser Facility, STFC Rutherford Appleton Laboratory, Harwell Oxford, Didcot OX11 0QX, UK*
42 *ISIS Neutron and Muon Source, STFC Rutherford Appleton Laboratory, Harwell Oxford, Didcot OX11 0QX, UK*
43 *Particle Physics Department, STFC Rutherford Appleton Laboratory, Harwell Oxford, Didcot, OX11 0QX, UK*
44 *Technology Department, STFC Daresbury Laboratory, Daresbury, Cheshire, WA4 4AD, UK*
45 *Department of Radiation Oncology, Stanford University, Stanford, CA 94305, USA*
46 *Department of Physics, SUPA, University of Strathclyde, Glasgow G4 0NG, UK*
47 *Department of Biomedical Sciences, Faculty of Science and Engineering, Swansea University, Singleton Park, Swansea, SA2 8PP, UK*
48 *Department of Physics, Faculty of Science and Engineering, Swansea University, Singleton Park, Swansea, SA2 8PP, UK*
49 *Department of Radiation Oncology, University of California San Francisco, San Francisco, CA, 94115, USA*
50 *University of Santiago de Compostela, Center for Research in Molecular Medicine and Chronic Diseases, 15782, Santiago de Compostela, Spain*